

Staff characteristics and school value added

Staff window: 2018-2024 | Observed histories: 2013 onward | 10 September 2026

Teacher credentials have the clearest association with VA. Orientador summary indices have weak links to both higher-education and test-score VA. Teacher links become much smaller after conditioning on school size and characteristics.

Balanced indices: correlations across all 12 outcomes

VA outcome	Orientador N	Orientador r	Teacher N	Teacher r
Math score	1,594	0.042	2,698	0.377
Language score	1,594	0.050	2,700	0.420
Admission-exam taking	1,642	0.054	2,706	0.334
Higher-ed enrollment	1,641	0.054	2,706	0.185
STEM enrollment	1,642	-0.009	2,706	-0.058
High-premium field	1,642	0.029	2,706	0.127
High-premium institution	1,642	-0.017	2,706	0.213
Projected income (full)	1,642	0.051	2,706	0.342
Projected income (field)	1,592	0.038	2,700	0.183
Projected income (institution)	1,592	0.037	2,700	0.144
Program accreditation years	1,592	0.032	2,700	0.115
Institution accreditation years	1,592	0.017	2,700	0.008

Each row uses saved empirical-Bayes (EB) VA and equally weighted schools. The balanced index gives equal weight to standardized experience and credential blocks. N is the number of schools with the index and that VA outcome.

The teacher **credentials block** alone correlates 0.406 with math VA, 0.512 with language VA and 0.446 with full projected-income VA. These are school-level associations, not causal effects of staff credentials or estimates of individual staff quality.

What the two scores measure

The common starting universe is 3,682 schools and 757,999 VA-sample students from grade-8 cohorts 2017-2020. The completed balanced indices cover **1,642 orientador schools** and **2,706 HS-teacher schools**. Missing scores remain missing.

Component	Orientadores	HS teachers
Experience: prior role years	Primary-function orientador years, 2013 through t-1	Observed HS-teaching years, 2013 through t-1
Experience: school-role spell	Consecutive years as orientador at this school, including t	Consecutive HS-teaching years at this school, including t
Credentials: university	Share with a reported university tertiary qualification	Same definition
Credentials: teaching title	Share with a reported teaching qualification	Share with a reported HS teaching qualification

Standardize the four school-period components within each role's complete-case sample. Average the two experience components and standardize that block. Do the same for credentials. Average the two standardized blocks, then standardize the balanced index to SD 1. All weights are independent of VA.

Roster, timing and missingness

Orientadores include primary or secondary function 9, across all appointments. Teachers require classroom function and their own regular youth-HS assignment. Count each person once per school-year. Later promotions never rewrite earlier histories. Observed experience is left-censored in 2013, not lifetime experience.

Average equally across staff within school-year and across active-role years within 2018-2024. Require seven known annual role counts, 80% observed eligible members for an annual attribute, and 80% valid active eligible years for its period mean. Indices additionally need at least three active years. No staff is not zero staff qualifications.

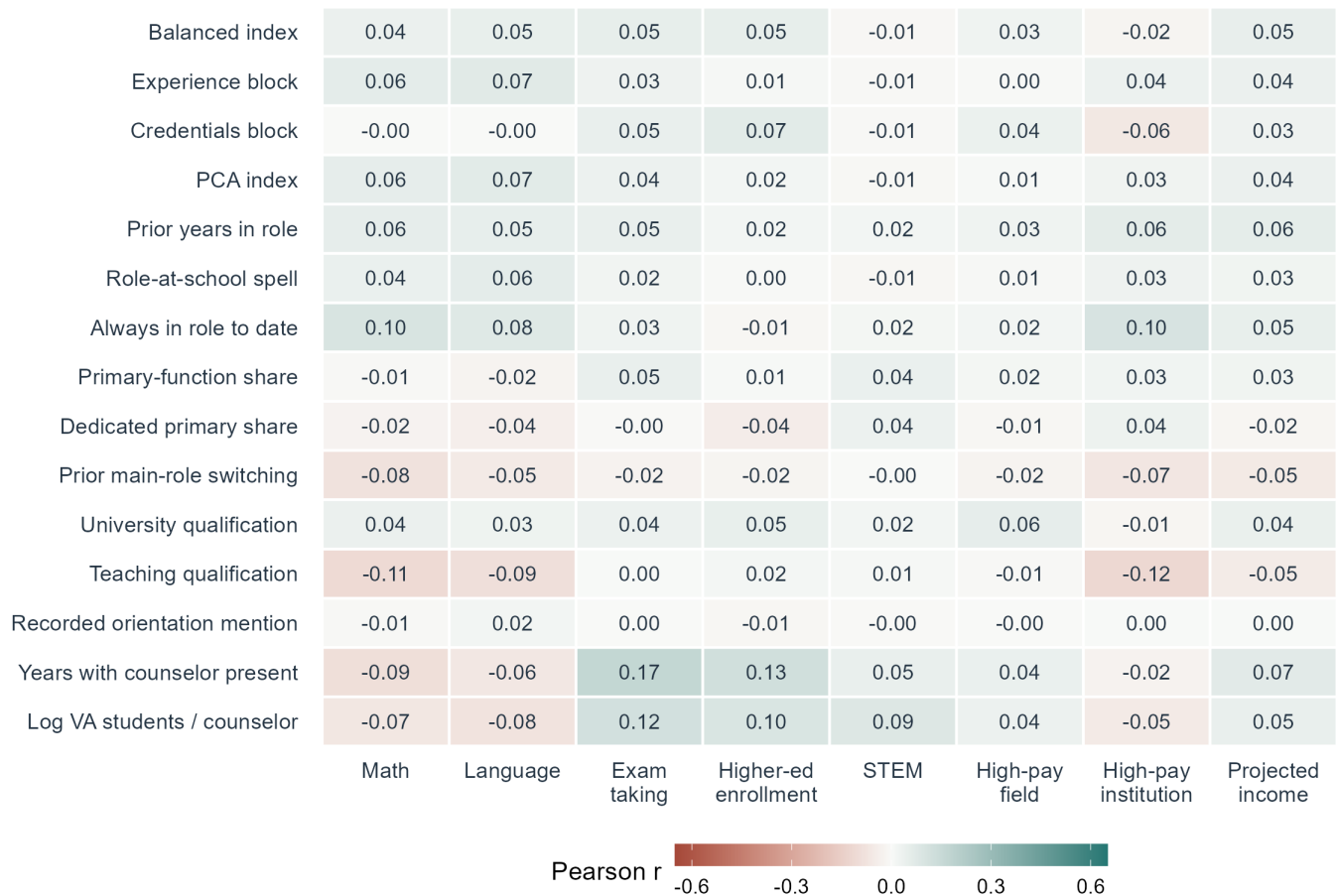
Fields that need separate treatment

Reported school tenure: zero-coded for 6.5% of orientador records in 2018 versus 95.1% in 2019, frequently for people already observed at that school. Keep reported tenure unchanged for audit; exclude its period average from the indices and analyze 2018 tenure separately.

Credentials: institution types are observed, not institution names or rankings. Orientation mentions are asked for applicable basic-education titles, so no recorded mention does not establish no counseling training. Math/language specialty matching uses the same HS subject-assignment slot.

Orientadores: weak summary-index links

Pearson correlations with EB school VA. Every displayed cell uses its own observed support; the complete table includes N, confidence intervals and missingness status.



The balanced index correlates about 0.04-0.05 with scores and full projected-income VA. None of its adjusted associations survives the role-wide multiple-testing correction. This does not establish that counseling is unimportant: the recorded characteristics may be noisy or incomplete.

Exploratory exceptions: 4 adjusted associations with $q < 0.05$

Characteristic	VA outcome	Adj. beta	BH q
Years with counselors present	Admission-exam taking	0.061	0.003
VA students / primary counselor	Admission-exam taking	-0.069	0.020
Counselors / 1,000 VA students	Math score	0.031	0.020
Prior main-role switching	High-premium institution	-0.045	0.044

These capacity/trajectory links are exploratory and do not validate a general orientador-quality score. Beta is in outcome SD per characteristic SD; q adjusts all 444 orientador metric-outcome tests.

HS teachers: credentials carry more of the association

Same color scale and outcome order as the orientador matrix. Subject matches refer to recorded teaching specialties among teachers assigned to that HS subject.



Teacher credentials have stronger correlations than the experience block. The balanced index is positively related to math, language and full projected-income VA, but it is not uniformly positively associated with every higher-education outcome.

Subject-specific credential checks

Pair	Schools	Pearson r	Adj. beta	BH q
Math match / math VA	2,956	0.163	0.042	0.015
Language match / language VA	2,961	0.193	0.067	<0.001

A teacher explicitly lacking a reported teaching title contributes zero recorded teaching specialty, rather than disappearing from the subject-matching denominator. Unknown credentials remain unknown. This does not equate non-teaching degrees with lack of subject knowledge.

How much remains after school adjustment?

Each row is a separate school-level regression. Both the staff measure and EB VA are standardized within its estimation sample. Controls are log VA-sample size and its square, administrative dependency, region, rural status, TP/artistic offerings, and basic-school offerings. School-directory controls are a 2024 snapshot.

Orientadores

Index/block	Outcome	N	Beta (SD)	95% HC1 CI	BH q
Balanced	Math	1,594	0.030	[-0.009, 0.070]	0.609
Balanced	Language	1,594	0.027	[-0.011, 0.065]	0.636
Balanced	Projected income	1,642	0.029	[-0.011, 0.069]	0.636
Experience	Math	1,594	0.026	[-0.013, 0.065]	0.653
Experience	Language	1,594	0.039	[0.002, 0.075]	0.470
Experience	Projected income	1,642	-0.004	[-0.043, 0.035]	0.964
Credentials	Math	1,594	0.017	[-0.020, 0.055]	0.741
Credentials	Language	1,594	-0.000	[-0.038, 0.037]	0.997
Credentials	Projected income	1,642	0.046	[0.007, 0.085]	0.376

HS teachers

Index/block	Outcome	N	Beta (SD)	95% HC1 CI	BH q
Balanced	Math	2,698	0.078	[0.044, 0.111]	<0.001
Balanced	Language	2,700	0.092	[0.061, 0.124]	<0.001
Balanced	Projected income	2,706	0.052	[0.017, 0.087]	0.017
Experience	Math	2,698	0.048	[0.017, 0.080]	0.015
Experience	Language	2,700	0.037	[0.007, 0.067]	0.057
Experience	Projected income	2,706	0.006	[-0.026, 0.037]	0.826
Credentials	Math	2,698	0.084	[0.050, 0.118]	<0.001
Credentials	Language	2,700	0.136	[0.099, 0.173]	<0.001
Credentials	Projected income	2,706	0.104	[0.062, 0.146]	<0.001

HC1 confidence intervals and t-based p-values are conditional on the saved estimated VA. They do not propagate VA-estimation uncertainty or address unobserved sorting. BH q-values are computed over all 444 orientador or 432 teacher tests, separately for adjusted associations.

PCA and robustness checks

PCA does not collapse experience and credentials equally

Component / diagnostic	Orientadores	HS teachers
Prior role years	0.701	0.650
School-role spell	0.698	0.617
University qualification	0.056	0.236
Teaching title (HS title for teachers)	0.137	0.375
PC1 share of four-component variance	42.9%	49.8%

For orientadores, PC1 is mostly observed experience. Experience and credentials blocks correlate only 0.062 for orientadores and 0.099 for teachers. Keep both blocks visible; the balanced index is a transparent summary, not evidence of one latent quality factor.

Balanced-index Pearson r under alternative checks

Role / outcome	Main	No private paid	100+ VA students	5+ active years	Unshrunk VA
Orient. / Math	0.042	0.062	0.036	0.028	0.058
Orient. / Language	0.050	0.064	0.051	0.037	0.066
Orient. / Projected income	0.051	0.056	0.045	0.052	0.054
Teacher / Math	0.377	0.235	0.412	0.385	0.365
Teacher / Language	0.420	0.311	0.454	0.430	0.407
Teacher / Projected income	0.342	0.224	0.356	0.345	0.352

The first three alternatives restrict schools. The last retains the matched sample and replaces EB VA with unshrunk VA. Full robustness tables retain sample sizes and adjusted estimates; rank and student-weighted correlations are included in the main CSV.

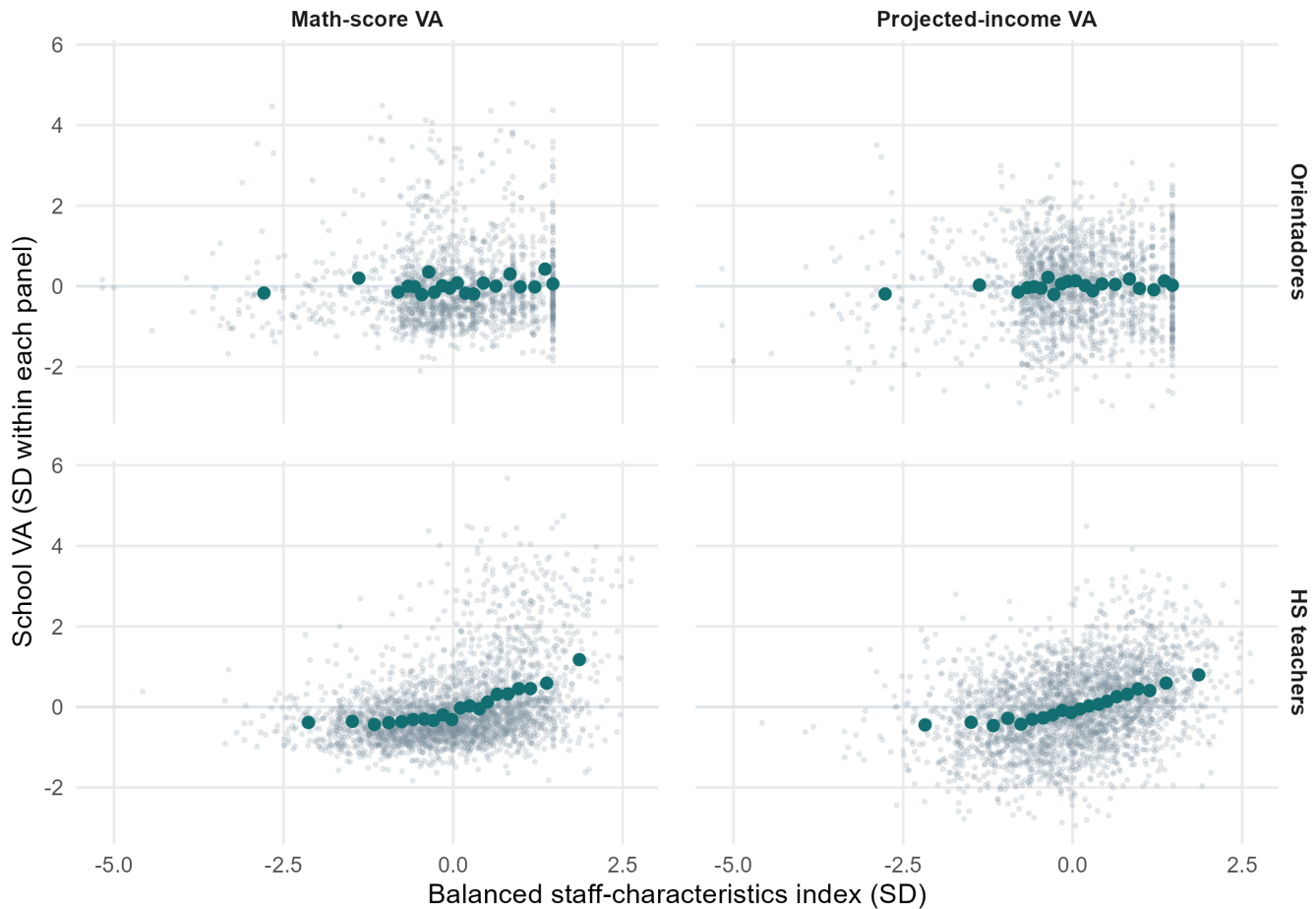
Early teacher-history sensitivity

VA outcome	Schools	Pearson r	Adjusted beta
Math	2,658	0.370	0.073
Language	2,660	0.414	0.089
Projected income	2,666	0.339	0.059

This separately fitted index starts prior HS experience in 2016 because 2013-2015 assignment slots are often missing. Teacher associations persist; it is not a reconstruction of the missing early history.

The school-level pattern

Balanced indices against math-score and full projected-income VA. Small gray points are schools; teal points are means within 20 rank-based index bins. Bins are descriptive, with no fitted causal curve or confidence band.



Interpretation

The teacher pattern is visible across the index distribution. The orientador pattern is much flatter. These figures use the balanced indices specified from staff variables, not weights selected to predict the displayed outcomes.

School adjustment substantially attenuates teacher associations. For example, the balanced teacher index has an unadjusted projected-income correlation of 0.342, versus an adjusted coefficient of 0.052 SD per index SD. The latter is not a 5.2% wage effect: its outcome is standardized estimated log projected-income VA.

Contemporaneous staff composition may reflect school sorting, management, resources or responses to past performance. A weak orientador correlation can also reflect measurement error, sparse staff rosters, and the incomplete measure of counseling-specific training. Neither pattern identifies the causal contribution of an individual staff member.

Methods, sources and verification

Analysis scope

The analysis crosses 37 orientador measures and 36 teacher measures with 12 saved outcomes: 876 main pairs. It also estimates 288 index-by-outcome robustness specifications across three school restrictions. All measured associations remain in the CSV outputs, including weak and negative estimates. Broken tenure-period and pure coverage/applicability diagnostics are retained outside the main association family.

Main correlations use equal school weights. Student-weighted correlations use the common broad-VA student counts, not inverse-variance weights. Pearson intervals use the usual correlation test; Spearman ranks canonicalize values to 10 significant digits to prevent floating-point tie breaking. Original precision is retained for indices, Pearson correlations and regressions.

Source data and definitions

Staff: MINEDUC annual public teacher directories in the project's Box raw-data folder, 2013-2024, with annual source paths, sizes and timestamps saved in the manifest. The 2024 ER_Cargos Docentes, bases publicas codebook documents tenure and qualifications; Annex VI identifies math subsector 32001 and Spanish-language 31001. Raw files are unchanged.

VA: the saved All-sample observational school VA and EB table. Outcome-specific supports are preserved; missing VA is not imputed and VA is not re-estimated. Higher-education enrollment has a slightly different source support, so its join is restricted to the common broad school universe.

School context: the official 2024 school directory. Staffing ratios use pooled VA-sample students divided by average annual staff headcount, not total HS enrollment, annual caseload or hours-adjusted capacity. Orientadores at mixed-level schools are not allocated exclusively to HS.

Checks completed

34 synthetic construction checks passed. Independent Python calculations verified 1,391,796 annual metric cells, 198,828 period aggregates, 7,014 index scores, all 876 Pearson/rank/weighted correlations and unshrunk-VA counterparts, main BH adjustments, and 12 exact HC1 regressions. Annual orientador counts reconcile to the existing staffing-ratio build. No analysis warnings were recorded.

Reusable outputs

Output	Contents
school_staff_indices_and_measures.csv	School-period components, indices and coverage
staff_va_correlations.csv	All 876 main associations, N, intervals, q-values
staff_va_index_robustness.csv	288 alternative-sample index associations
Index standardizations and PCA loadings	Reproducible scaling, weights and PCA diagnostics

Tables and audits are under data/clean/staff_quality_va/. The task README documents the full build and refresh sequence. Individual staff records remain local and excluded from Git. Remaining data limitations are recorded in project memory; the analysis does not resolve the missing early teaching assignments or the reported-tenure coding break.